ChE 210 Homework Set #10 Due on Friday, 21 November 2025

(1) The standard heat of the reaction

$$CaC2(s) + 5\ H2O(l) \rightarrow.\ CaO(s) + 2\ CO2(g) + 5\ H2(g)$$

is $\Delta\hat{H}_r^\circ = +69.36\,\text{kJ/mol}$.

(a) Is the reaction exothermic or endothermic at 25°C? Would you have to heat or cool the reactor to keep the temperature constant? What would the temperature do if the reactor ran adiabatically? What can you infer about the energy required to break the molecular bonds of the reactants and that released when the product bonds are formed?

(b) Calculate $\Delta\hat{U}_r^\circ$ for this reaction. Briefly explain the physical significance of your calculated value.

(c) Suppose you charge 150.0 g of CaC2 and liquid water into a rigid container at 25°C, heat the container until the calcium carbide reacts completely, and cool the products back down to 25°C, condensing essentially all the unconsumed water. Write and simplify the energy balance equation for this closed constant-volume system and use it to determine the net amount of heat (kJ) that must be transferred to (or from) the reactor. Indicate the direction of heat transfer.

(2) Murphy 6.32 (N.B. Consider the change in energy in the inner combustion chamber as being completely due to the chemical reaction and ignore the sensible energy change of the $C_8H_{10}O_3$ and its products.)

(3) Murphy 6.34

(4) Use tabulated heats of formation to determine the standard heats of the following reactions in kJ/mol, letting the stoichiometric coefficient of the first reactant in each reaction equal one.

(a) Nitrogen (N_2) + oxygen (O_2) react to form nitric oxide (NO) al in the gas state.

(b) Gaseous *n*-pentane + oxygen react to form carbon monoxide + liquid water.

(c) Liquid *n*-hexane + oxygen react to form carbon dioxide + water vapor. After doing the calculation, write the stoichiometric equations for the formation of the reactant and product species, then use Hess's law to derive the formula you used to calculate $\Delta\hat{H}_r^\circ$.

(d) Liquid sodium sulfate + carbon monoxide react to form liquid sodium sulfide + carbon dioxide. [The sodium compounds are not listed in the textbook tables. You will need to find data elsewhere.]

(5) Murphy 6.37

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1.

$$CaC_{2(s)} + 5H_2O_{(l)} \longrightarrow CaO_{(s)} + 2CO_{2(g)} + 5H_{2(g)}$$ ✓

(A)

1. The reaction is endothermic because the standard heat of reaction is positive. This means the reaction absorbs heat from the surroundings, which makes it an endothermic reaction.
2. To keep the temperature of the reactor constant throughout the reaction, heat must be supplied to the reactor. Since the reaction is continuously taking in heat, if no heat is added it would cause the reactors temperature to decrease.
3. In an adiabatic reactor, the reaction will absorb heat in order to proceed. However, because no heat can be transferred from the surroundings into the reactor system. The reactors temperature will decrease as the reaction consumes the thermal energy.
4. More energy is required to overcome the forces between the reactants than the energy released when the bonds in the products are formed. Therefore, the net result is an absorption of energy.

When GAS reacts! (Δn of gas mols only) ✓

(B) $\Delta\hat{U}_r^\circ = ?$ $\quad \Delta\hat{H}_r^\circ = \Delta\hat{U}_r^\circ + RT\Delta n$ $\quad \left(8.3144\frac{J}{gmol\cdot K}\right)\left(\frac{1 kJ}{10^3 J}\right) = 0.0083144\frac{kJ}{gmol\cdot K}$ $\quad \hat{H} = \hat{U} + P\hat{V}$ ✓

$$\Delta\hat{U}_r^\circ = \Delta\hat{H}_r^\circ - RT\Delta n = \left(69.36\frac{kJ}{gmol}\right) - \left(0.0083144\frac{kJ}{gmol\cdot K}\right)(7 - 0\,gmol)(298K) = \boxed{52.017\frac{kJ}{gmol}}$$ ✓

(C) $\boxed{\text{Rigid Tank Reactor } (CaC_{2(s)} + H_2O_{(l)})}$

$$\frac{150g\,CaC_2 \mid 1\,gmol\,CaC_2}{\mid 64.1\,g\,CaC_2} = 2.34\text{ gmols } CaC_2 \text{ reacts w/ } 11.7 \text{ gmol } H_2O$$

Eng Model
- Closed System
- Rigid Tank (Constant Volume)
- Ideal Gas
- Kinetic/Potential Energy Negligible.
- Well Mixed

Closed System

$\Delta E = Q + W$

$\Delta U + \cancel{\Delta E_k} + \cancel{\Delta E_p} = Q + \cancel{W}$

$\Delta U = Q$

$n(\Delta\hat{U}_r^\circ) = Q$ ✓

↳ for standard condition of 1 mole CaC_2 reacting.

↳ Scale Up to 2.34 gmol or 150g of CaC_2 reacting.

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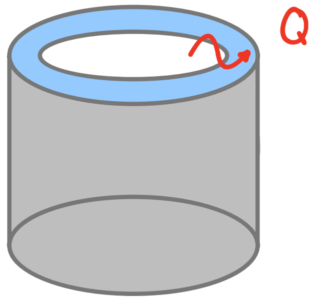
$$Q = (2.34\text{ gmols})\left(52.017\frac{kJ}{gmol}\right) = \boxed{121.7198\text{ kJ}}$$ ✓

2.

P6.32 In a bomb calorimeter, a sample is placed into an inner combustion chamber and combusted to CO_2 and H_2O. The combustion chamber sits inside another chamber which is filled with an exact amount of water. The temperature of the outer chamber is carefully monitored. In one experiment, 2.7 g of a proprietary organic chemical with the molecular formula $C_8H_{10}O_3$ is placed in a bomb calorimeter and combusted. In the outer chamber, which contains 650 g water, the temperature increases from 23.6°C to 36.2°C. Use these data to calculate an enthalpy of formation (kJ/g) for the unknown chemical.

$2.7g\ C_8H_{10}O_3$
$650g\ H_2O$
$T_f = 36.2\ C°$
$T_i = 23.6\ C°$



$$2.7g\ C_8H_{10}O_3 \times \frac{1\ gmol\ C_8H_{10}O_3}{157.17\ g\ C_8H_{10}O_3} = $$
$$= 0.017513\ gmols\ C_8H_{10}O_3$$ ✓

$$C_8H_{10}O_3 + 9O_2 \longrightarrow 8CO_2 + 5H_2O$$ ✓

Energy Balance for Water

$$\Delta U + \cancel{\Delta E_K} + \cancel{\Delta E_p} = Q + \cancel{W}$$

$$Q = mc_p\Delta T = \left(650g \cdot \frac{1gmol}{18.01\,gH_2O}\right)(75.4\ J/gmol\cdot K)(36.2-23.6)K$$

$$\Delta U = Q$$

$$Q = mc\Delta T = (650g)(4.18\ J/g\cdot C°)(36.2-23.6)C° = \boxed{34234\ J}$$ (+: Heat to Water)

Energy Balance for Chemical

$$\Delta U + \cancel{\Delta E_K} + \cancel{\Delta E_p} = Q + \cancel{W}$$

$\Delta U = Q = -34234J$ (For whole rxn of 2.7g or 0.017513 gmols $C_8H_{10}O_3$)

$$\Delta U = (\text{mols of rxn})(\Delta\hat{U}_r) \quad \Delta\hat{U}_r = \frac{(\text{mols of rxn})(\Delta\hat{U}_r)}{(\text{mols of rxn})} \quad \frac{(\text{mols of rxn})}{(\text{mols of rxn})} = 1$$

— To represent standard rxn of 1 mole of chemical.

$$\Delta\hat{U}_r = \frac{-34234J}{0.017513\ gmols\ C_8H_{10}O_3} = -1954773\frac{J}{gmol} \text{ or } \boxed{-1955\frac{kJ}{gmol}}$$ ✓

Enthalpy of Reaction

gas $n_{reactants}$ — gas $n_{products}$

$$\Delta\hat{H}_r = \Delta\hat{U}_r + RT\Delta n_r = -1955\frac{kJ}{gmol} + \left(0.0083144\frac{kJ}{gmol\cdot K}\right)(8-9\ gmol)(298K) = -1957\frac{kJ}{gmol}$$ ✓

Enthalpy of Formation for Chemical

$$\Delta\hat{H}_r^\circ = \sum_i (\nu_i)(\Delta\hat{H}_{f,i}^\circ)$$ ✓

$$-1957\frac{kJ}{gmol} = (-1)(y) + (-9)(0) + (8)\left(-393.5\frac{kJ}{gmol}\right) + (5)\left(-285.84\frac{kJ}{gmol}\right)$$

$$y = -2620\frac{kJ}{gmol} = \Delta\hat{H}_{f,C_8H_{10}O_3}$$

$$\Delta H_{f,C_8H_{10}O_3} = -2620\frac{kJ}{gmol}\left(\frac{1gmol\ C_8H_{10}O_3}{154.17g\ C_8H_{10}O_3}\right) = \boxed{16.99\frac{kJ}{g}}$$ ✓

20/20.

3.

P6.34 Pocket-sized hot packs are popular as handwarmers with skiers, other winter-sports enthusiasts, and people who need to work outside in the cold. With one type of disposable hotpack, Fe filings are placed in a sealed bag. When the seal is broken by the user, air leaks in. Oxygen reacts with the iron, producing iron oxides, mainly Fe_2O_3. Marketing studies suggest that consumers want a slow heat release of about 100 kJ/h and would like the hotpacks to last about 6 h. How much Fe would you put in the hot pack? How much would the mass of the hotpack change over the course of its use (total g and %)?

$$2Fe + \frac{3}{2}O_2 \longrightarrow Fe_2O_3$$

(A) How much Fe would go in hot pocket?

$$\Delta \hat{H}_r^\circ = \sum_i (\nu_i)(\Delta \hat{H}_{f,i}^\circ) = (-2)(0) + (-\tfrac{3}{2})(0) + (1)(-830.5\tfrac{kJ}{gmol}) = \boxed{-830.5\tfrac{kJ}{gmol}}$$

Desired heat released is 600 kJ. If $-830.5\frac{kJ}{gmol}$ is released for every 2 mols Fe that react.

$$\left(600\ kJ \Big/ \frac{-830.5\frac{kJ}{gmol}}{2\ gmol\ Fe}\right) = \boxed{1.44\ gmol\ of\ Fe}$$ needed to react to reach desired amount of heat.

$$1.44\ gmol\ Fe \times \left(\frac{56\ g\ Fe}{1\ gmol\ Fe}\right) = \boxed{80\ g\ of\ Fe\ in\ hot\ pocket}$$

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(B) How will mass of hot pocket change while it is being used?

$$\frac{1.44\ gmol\ Fe}{} \left| \frac{1\ gmol\ Fe_2O_3}{2\ gmol\ Fe} \right| \frac{160\ g\ Fe_2O_3}{1\ gmol\ Fe_2O_3} = 115\ g\ of\ Fe_2O_3 \Longrightarrow$$ product in hot pack after reaction

$m_f = 115g$
$m_i = 80g$

$$\Delta m = (m_f - m_i) = \boxed{35g}$$

$$\frac{m_f - m_i}{m_i} \times 100 = \frac{115-80}{80} \times 100 = \boxed{43.75\%}$$

(4) Use tabulated heats of formation to determine the standard heats of the following reactions in kJ/mol, letting the stoichiometric coefficient of the first reactant in each reaction equal one.

(a) Nitrogen (N_2) + oxygen (O_2) react to form nitric oxide (NO) al in the gas state.

(b) Gaseous *n*-pentane + oxygen react to form carbon monoxide + liquid water.

(c) Liquid *n*-hexane + oxygen react to form carbon dioxide + water vapor. After doing the calculation, write the stoichiometric equations for the formation of the reactant and product species, then use Hess's law to derive the formula you used to calculate $\Delta\hat{H}_r^\circ$.

(d) Liquid sodium sulfate + carbon monoxide react to form liquid sodium sulfide + carbon dioxide. [The sodium compounds are not listed in the textbook tables. You will need to find data elsewhere.] NIST/2018OOK

4.

Heat of Formation

$$\Delta\hat{H}_r^\circ = \sum_i (\nu_i)(\Delta\hat{H}_{f,i}^\circ)$$

Heats of Combustion

$$\Delta\hat{H}_r^\circ = -\sum_i (\nu_i)(\Delta\hat{H}_{c,i}^\circ)$$

(A) $N_{2(g)} + O_{2(g)} \longrightarrow 2NO_{(g)}$ (Table B.3)

$$\Delta\hat{H}_r^\circ = (-1)(0) + (-1)(0) + (2)\left(90.25\tfrac{kJ}{gmol}\right) = \boxed{180.5\tfrac{kJ}{gmol}}$$

✓

(B) $C_5H_{12(g)} + \tfrac{11}{2}O_{2(g)} \longrightarrow 5CO_{(g)} + 6H_2O_{(l)}$ (Table B.3)

$$\Delta\hat{H}_r^\circ = (-1)\left(-146.76\tfrac{kJ}{gmol}\right) + \left(-\tfrac{11}{2}\right)(0) + (5)\left(-110.53\tfrac{kJ}{gmol}\right) + (6)\left(-285.84\tfrac{kJ}{gmol}\right) = \boxed{-2120.93\tfrac{kJ}{gmol}}$$

✓

(C) $C_6H_{14(l)} + \tfrac{19}{2}O_{2(g)} \longrightarrow 6CO_{2(g)} + 7H_2O_{(g)}$ (Table B.3)

$$\Delta\hat{H}_r^\circ = (-1)\left(-198.8\tfrac{kJ}{gmol}\right) + \left(-\tfrac{19}{2}\right)(0) + (6)\left(-393.5\tfrac{kJ}{gmol}\right) + (7)\left(-241.83\tfrac{kJ}{gmol}\right) = \boxed{-3855.01\tfrac{kJ}{gmol}}$$

✓

Hess's Law

C_6H_{14}:	$6C + 7H \longrightarrow C_6H_{14}$	$\Delta\hat{H}_r^\circ = -198.8$
O_2:	$O_2 \longrightarrow O_2$	$\Delta\hat{H}_r^\circ = 0$
CO_2:	$C + O_2 \longrightarrow CO_2$	$\Delta\hat{H}_r^\circ = -393.5$
H_2O:	$H_2 + \tfrac{1}{2}O_2 \longrightarrow H_2O$	$\Delta\hat{H}_r^\circ = -241.83$

$C_6H_{14} \longrightarrow 6C + 7H$	$-1\cdot\left(-198\tfrac{kJ}{gmol}\right)$
$\tfrac{19}{2}[O_2 \longrightarrow O_2]$	$\tfrac{19}{2}\cdot(0)$
$6[C + O_2 \longrightarrow CO_2]$	$6\cdot\left(-393.5\tfrac{kJ}{gmol}\right)$
$7[H_2 + \tfrac{1}{2}O_2 \longrightarrow H_2O]$	$+\ 7\cdot\left(-241.83\tfrac{kJ}{gmol}\right)$
	$-3855.01\tfrac{kJ}{gmol}$

(D) $Na_2SO_{4(l)} + 4CO_{(g)} \longrightarrow Na_2S_{(l)} + 4CO_{2(g)}$ (NIST)

+24.3

$$\Delta\hat{H}_r^\circ = (-1)\left(-1356.38\tfrac{kJ}{gmol}\right) + (-4)\left(-110.53\tfrac{kJ}{gmol}\right) + (-1)\left(-323.94\tfrac{kJ}{gmol}\right) + (-4)\left(-393.51\tfrac{kJ}{gmol}\right)$$

+6.7

$$\Delta\hat{H}_r^\circ = \boxed{-99.48\tfrac{kJ}{gmol}}$$

✓

$\frac{20}{20}$

5.

P6.37 A pure sodium crystal (100 g) sits in a steel cup. One drop (1 mL) water is added. Hydrogen gas is produced. What else happens? Explain, as quantitatively as possible, why it is dangerous to add water to pure metals like sodium.

$$Na + H_2O \longrightarrow \frac{1}{2}H_2 + NaOH$$

$$\Delta \hat{H}_r^\circ = \sum_i (\nu_i)(\Delta \hat{H}^\circ_{f,i}) = (-1)(0) + (-1)\left(-285.84\frac{kJ}{gmol}\right) + \left(+\frac{1}{2}\right)(0) + (+1)\left(-425.69\frac{kJ}{gmol}\right) = -139.85\frac{kJ}{gmol}$$

$$\frac{1\,mL\,H_2O \;|\; 1\,g\,H_2O \;|\; 1\,gmol\,H_2O}{\;|\; 1\,mL\,H_2O \;|\; 18.01\,g\,H_2O} = 0.056\ gmol\ H_2O \times \left(\frac{1\,mol\,Na}{1\,mol\,H_2O}\right) = 0.056\ gmol\ Na$$

$$\frac{100\,g\,Na \;|\; 1\,gmol\,Na}{\;|\; 22.98\,g\,Na} = 4.35\ gmol\ Na.$$

0.056 gmol Na reacts w/ H_2O

4.29 gmol Na remains unreacted.

$$\Delta \hat{H}_r(\text{mols of Na reacting}) = \left(139.85\frac{kJ}{gmol}\right)(0.056\,gmol) = 7.84\ kJ$$

$(Q = \Delta U)$ of Na

$7.84\ kJ = mc\Delta T$

$$\Delta T = \frac{7.84\,kJ}{\left(0.02823\frac{kJ}{mol\cdot K}\right)(4.29\,gmol)} = 65\ K \text{ or } C^\circ$$

The reaction could raise the temperature of sodium crystal by 65 degrees. Which is a great temperature difference based on the amount of sodium that reacted.

$\frac{20}{20}$